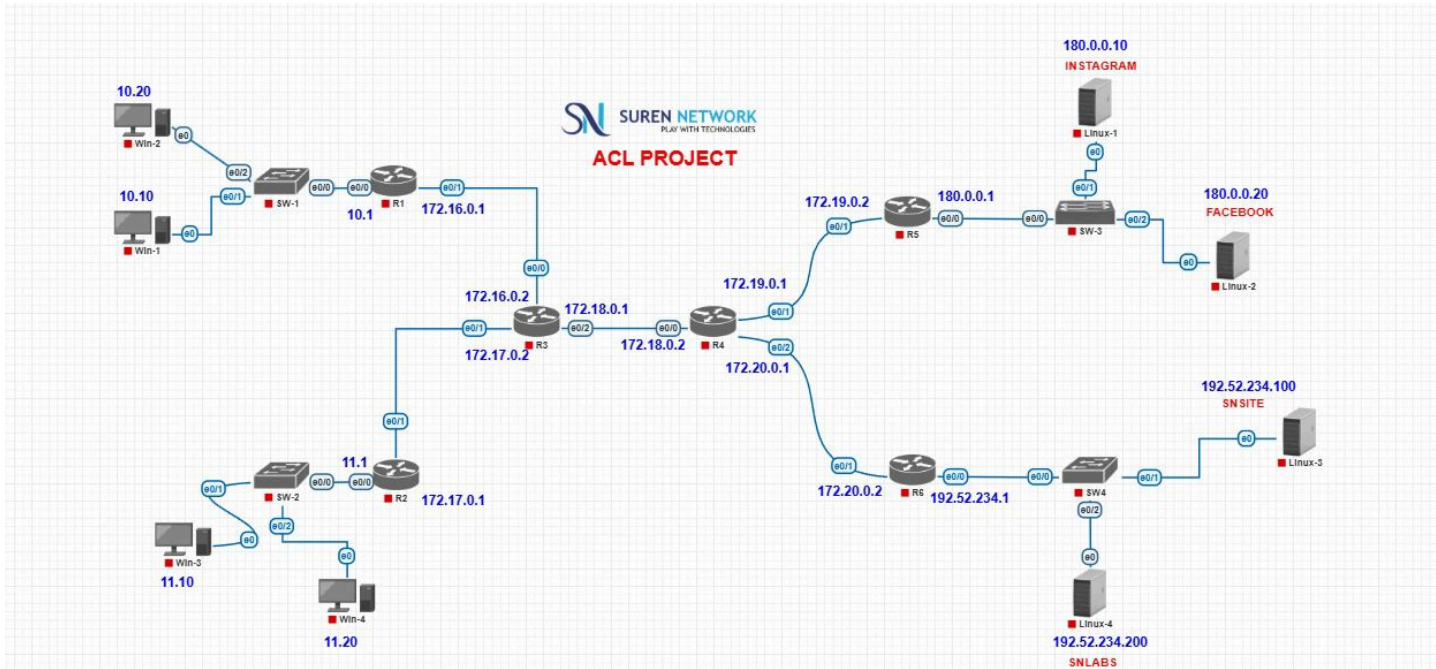






ACL PROJECT

Scenario 1 : Allow 10.0.0.0 to communicate only instagram server using port 80

Scenario 2 : Allow ping communication from 10.0.0.10 to instagram server

Scenario 3 : Allow 10.0.0.20 to communicate facebook server using ssh & http

Scenario 4 : Allow 10.0.0.0 to communicate snsite server using http

Scenario 5 : Allow 10.0.0.10 to communicate snlabs server using http & icmp

Scenario 6 : Allow 11.0.0.0 to communicate instagram & facebook server using http

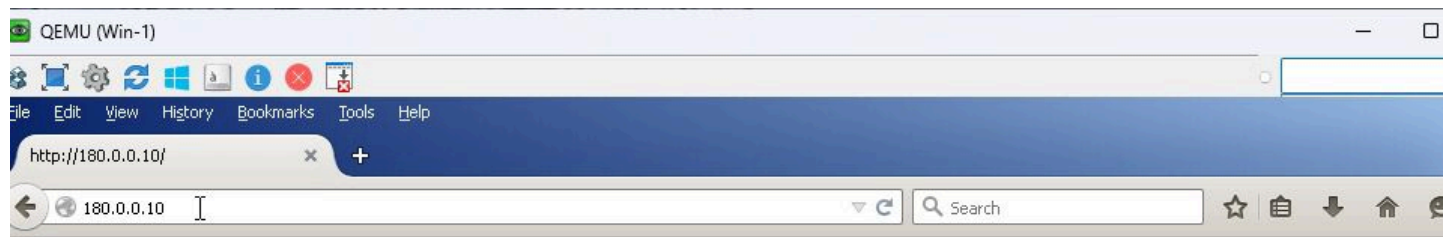
Scenario 7 : Allow 11.0.0.10 to ping facebook server

Scenario 8 : Allow 11.0.0.10 to ping snlab server

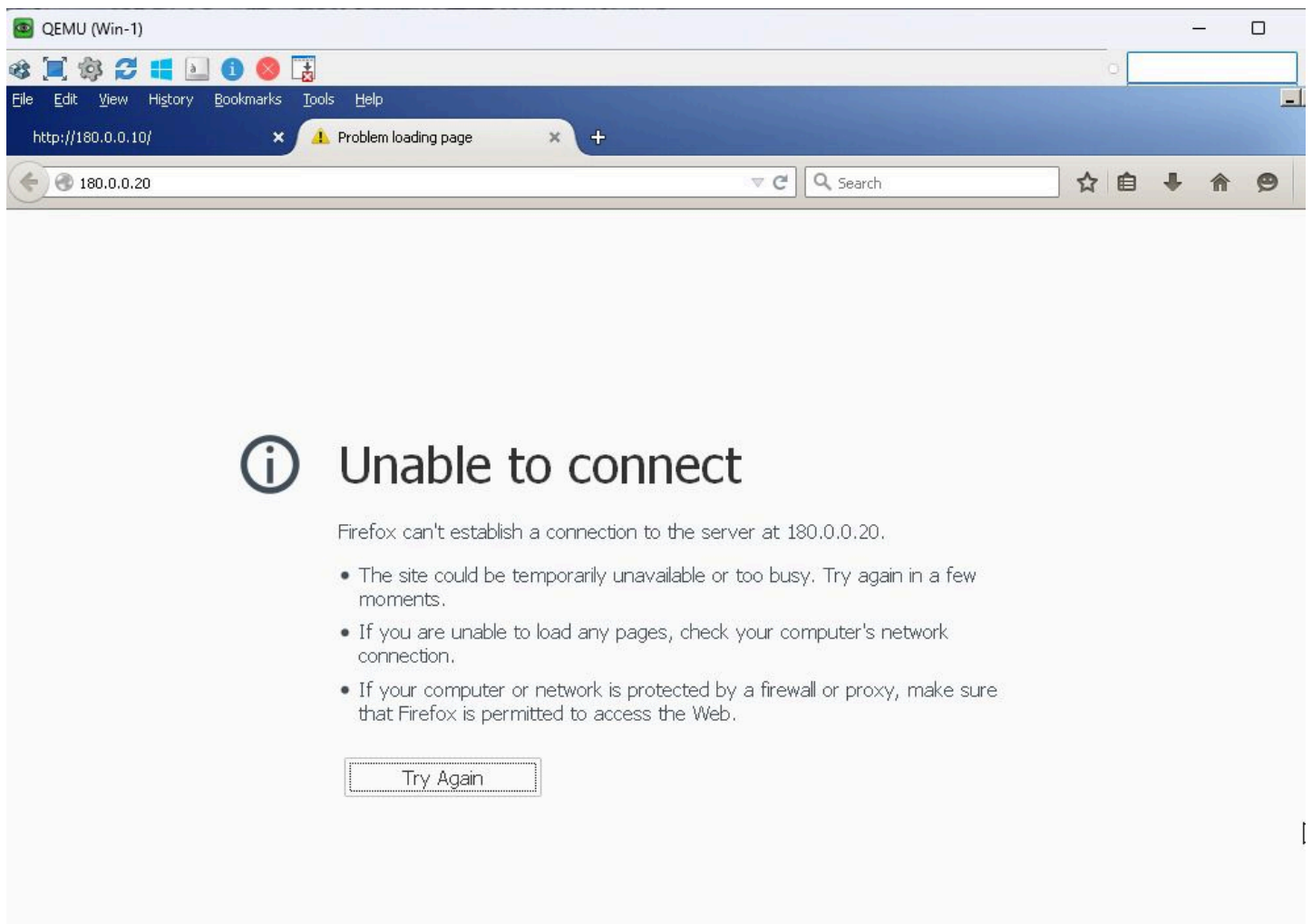
Scenario 9 : Allow 11.0.0.0 to access snlab server

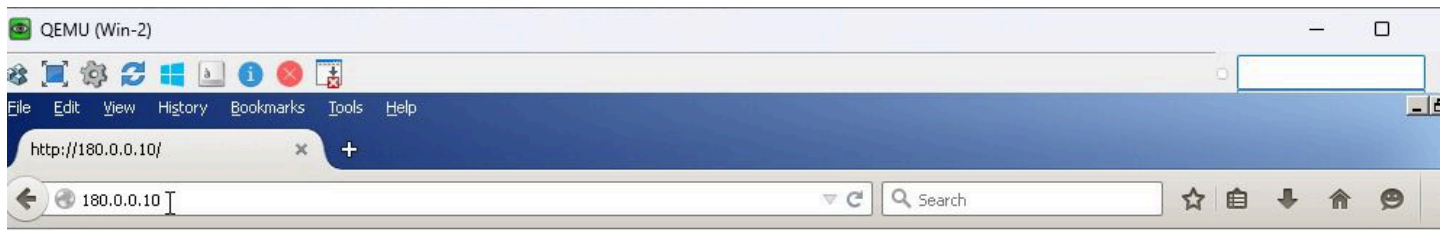
Scenario 10 : Deny 11.0.0.0 to access 10.0.0.0

```
R3
!
interface Ethernet0/0
 ip address 172.16.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/1
 ip address 172.17.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/2
 ip address 172.18.0.1 255.255.0.0
 ip access-group 1 in
 duplex auto
!
interface Ethernet0/3
 no ip address
 ip access-group 100 in
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 180.0.0.10 eq www
!
control-plane
!
!
--More--
```

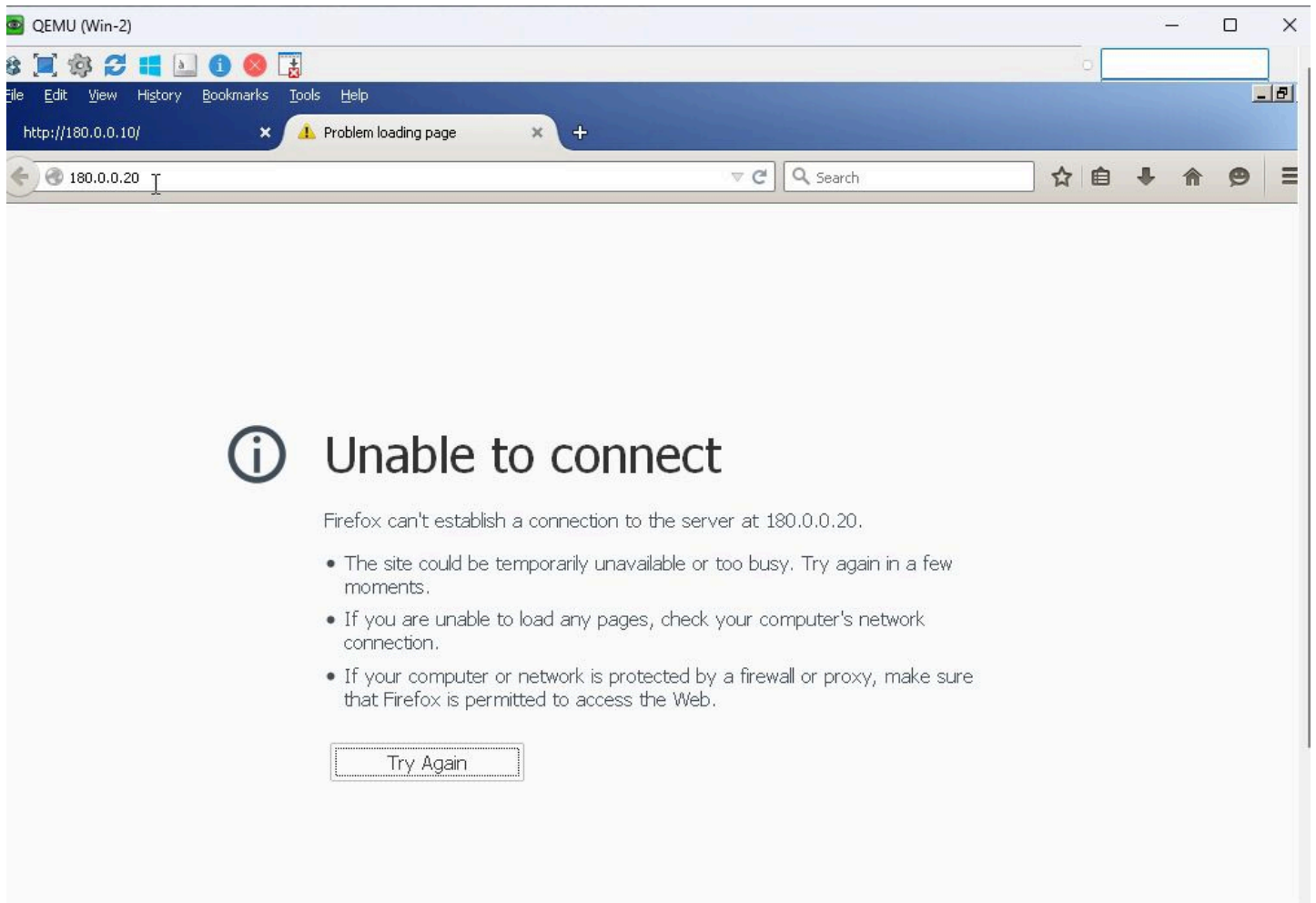


Instagram server - 180.0.0.10





Instagram server - 180.0.0.10



```
R3#
R3#
R3#
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#access-list 100 permit icmp 10.0.0.0 0.0.0.0 180.0.0.10 0.0.0.0
R3(config)#end
R3#
*Aug  7 16:41:05.589: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#
R3#
R3#
R3#
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#$ 100 permit icmp 10.0.0.10 0.0.0.0 180.0.0.10 0.0.0.0 echo
R3(config)#$ 100 permit icmp 10.0.0.10 0.0.0.0 180.0.0.10 0.0.0.0 echo-reply
R3(config)#end
R3#
*Aug  7 16:42:24.726: %SYS-5-CONFIG_I: Configured from console by console
R3#
```



```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 180.0.0.10

Pinging 180.0.0.10 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 180.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 180.0.0.10

Pinging 180.0.0.10 with 32 bytes of data:

Reply from 180.0.0.10: bytes=32 time=6ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60
Reply from 180.0.0.10: bytes=32 time=5ms TTL=60

Ping statistics for 180.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 4ms

C:\Documents and Settings\user>ping 180.0.0.20

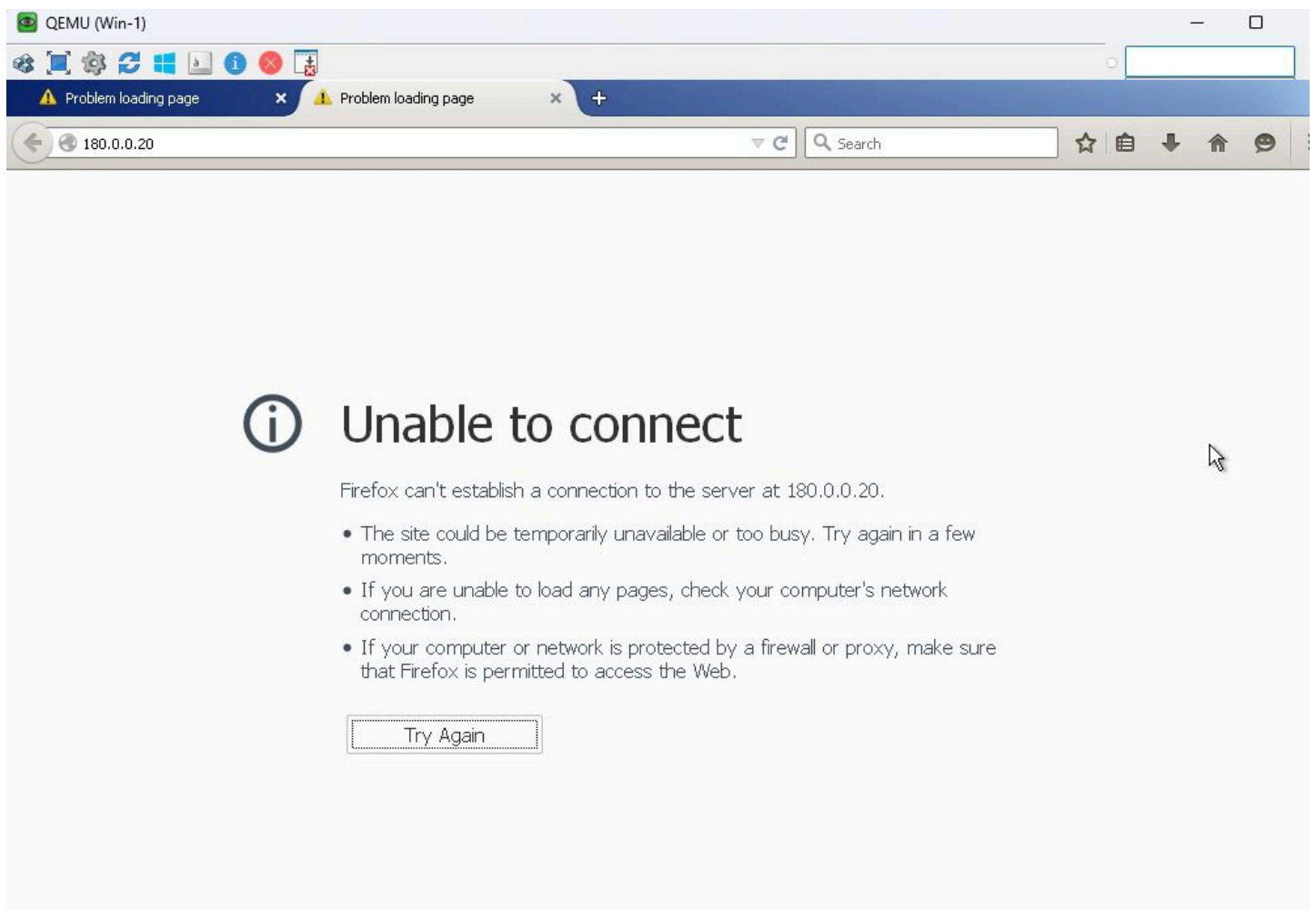
Pinging 180.0.0.20 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 180.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

```
R3
interface Ethernet0/1
 ip address 172.17.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/2
 ip address 172.18.0.1 255.255.0.0
 duplex auto
!
interface Ethernet0/3
 no ip address
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp host 10.0.0.20 host 180.0.0.20 eq www
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 0 0
--More--
```



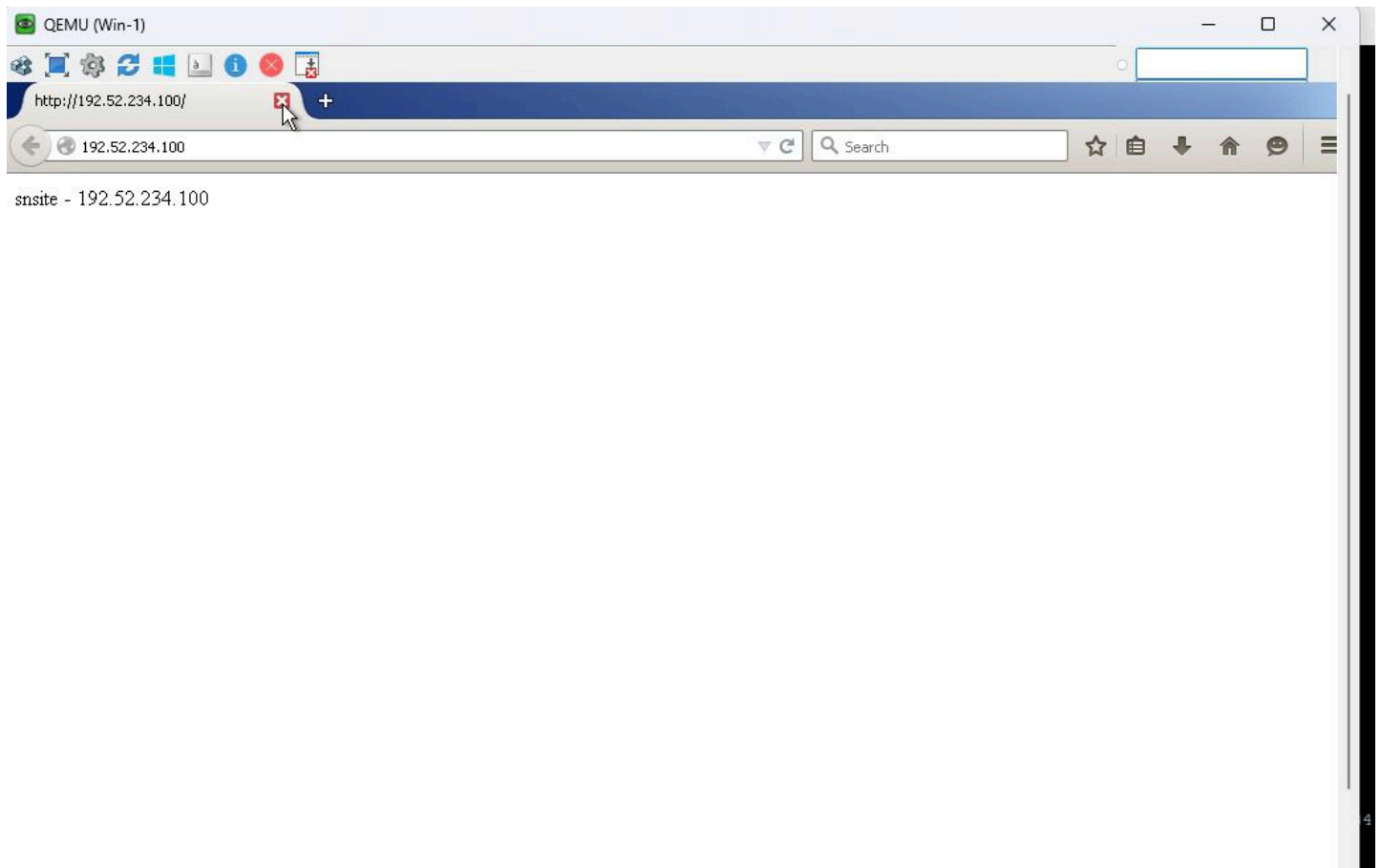


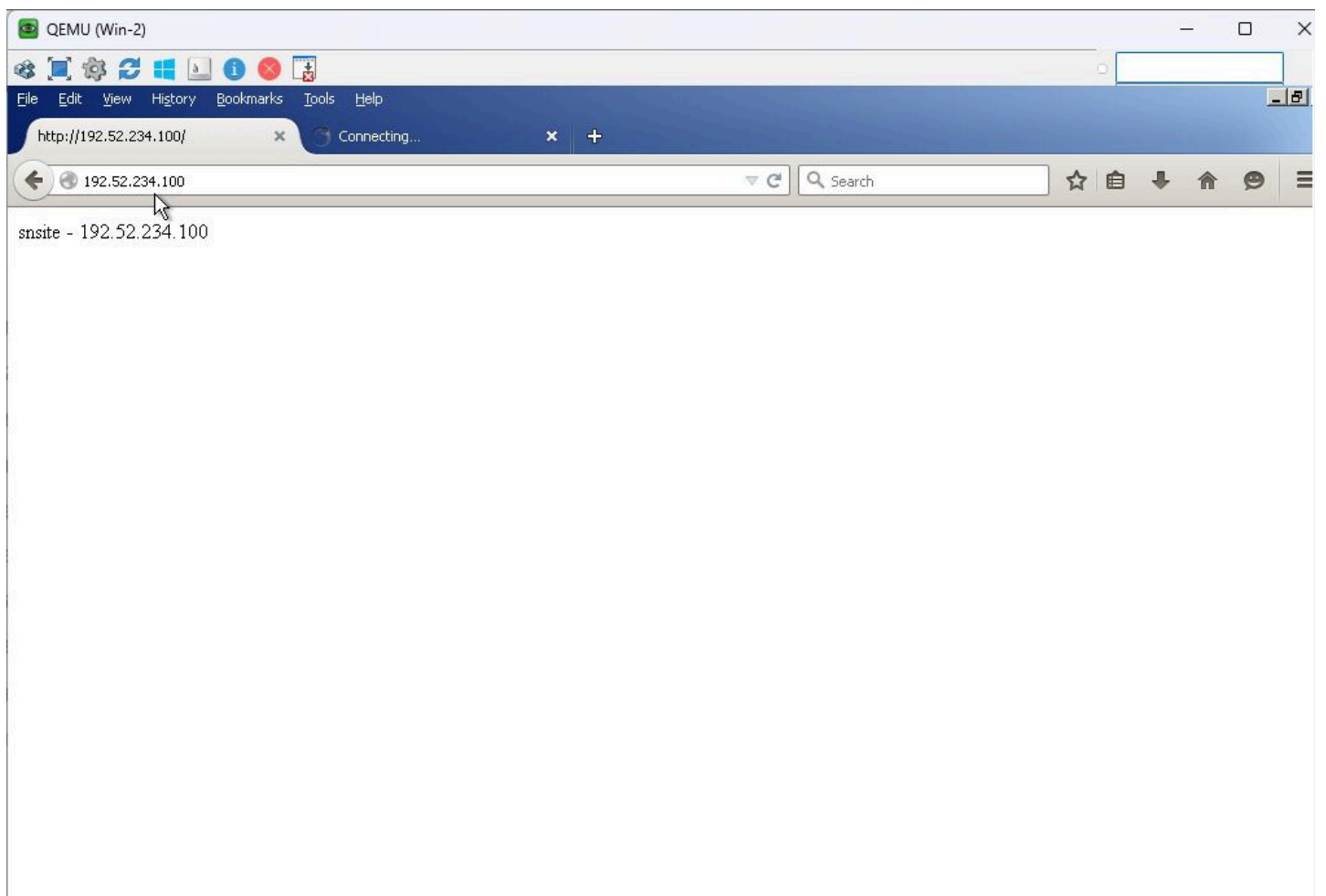
facebook server -180.0.0.20



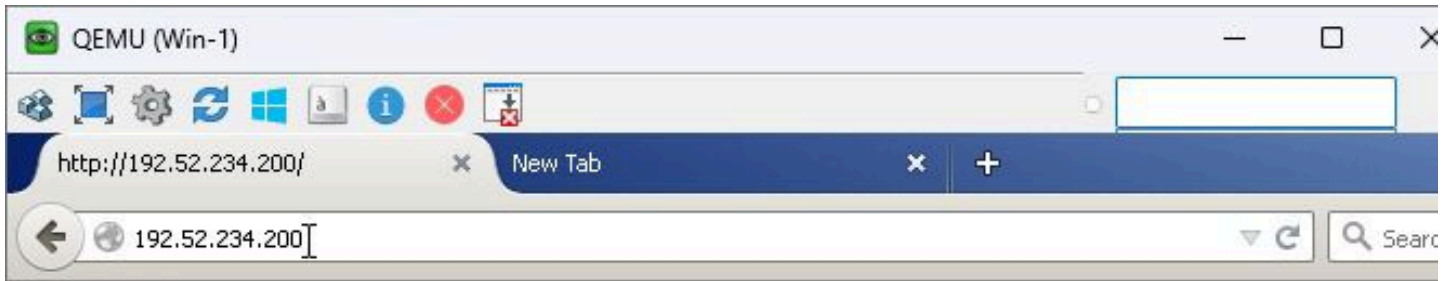
R3

```
!
interface Ethernet0/0
 ip address 172.16.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/1
 ip address 172.17.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/2
 ip address 172.18.0.1 255.255.0.0
 ip access-group 1 in
 duplex auto
!
interface Ethernet0/3
 no ip address
 ip access-group 100 in
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
!
control-plane
!
```

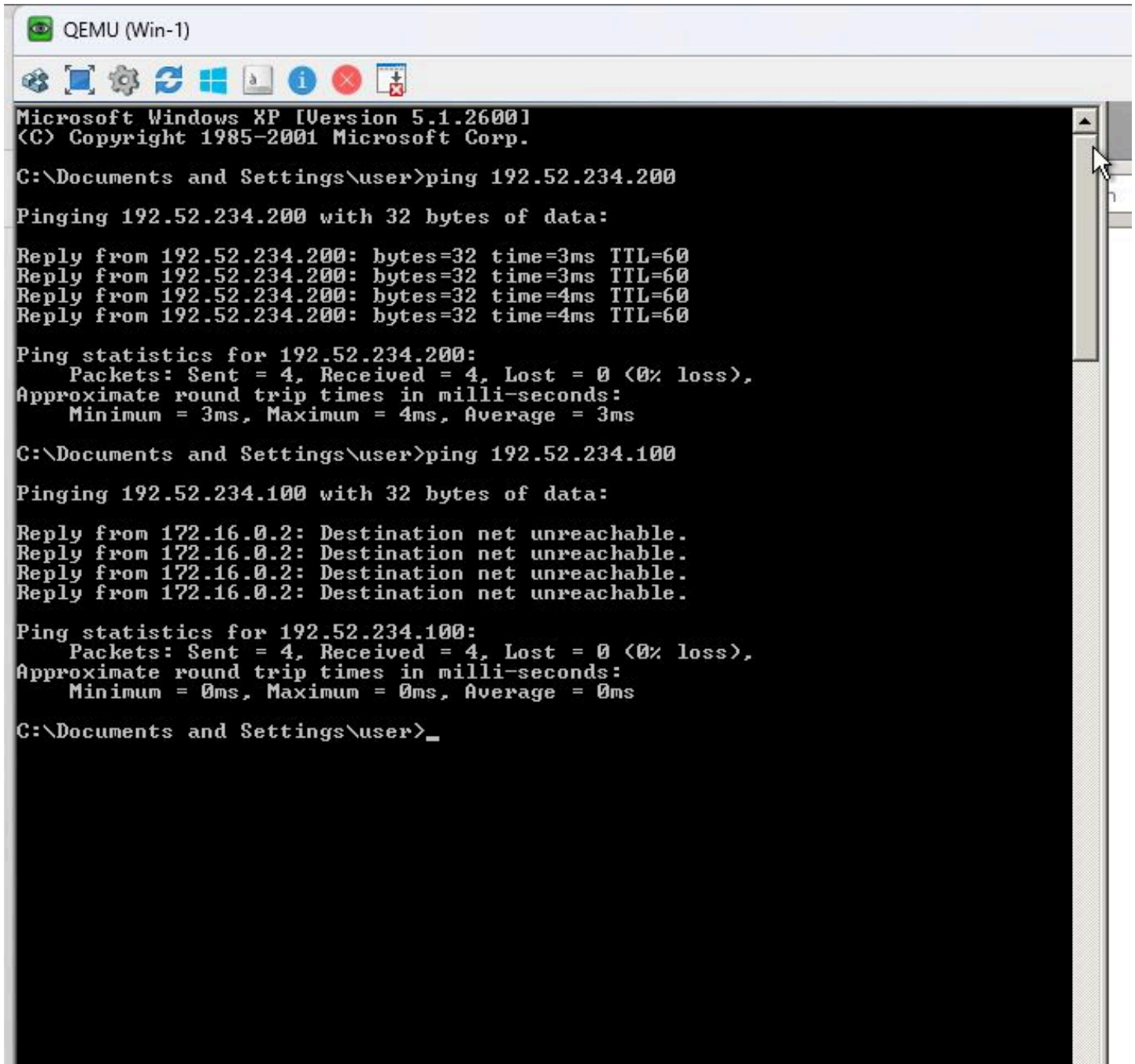





```
R3
interface Ethernet0/0
 ip address 172.16.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/1
 ip address 172.17.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/2
 ip address 172.18.0.1 255.255.0.0
 ip access-group 1 in
 duplex auto
!
interface Ethernet0/3
 no ip address
 ip access-group 100 in
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
access-list 100 permit tcp host 10.0.0.10 host 192.52.234.200 eq www
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo-reply
!
control-plane
--More--
```

sn labs - 192.52.234.200



```
QEMU (Win-1)
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 192.52.234.200

Pinging 192.52.234.200 with 32 bytes of data:

Reply from 192.52.234.200: bytes=32 time=3ms TTL=60
Reply from 192.52.234.200: bytes=32 time=3ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60

Ping statistics for 192.52.234.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>ping 192.52.234.100

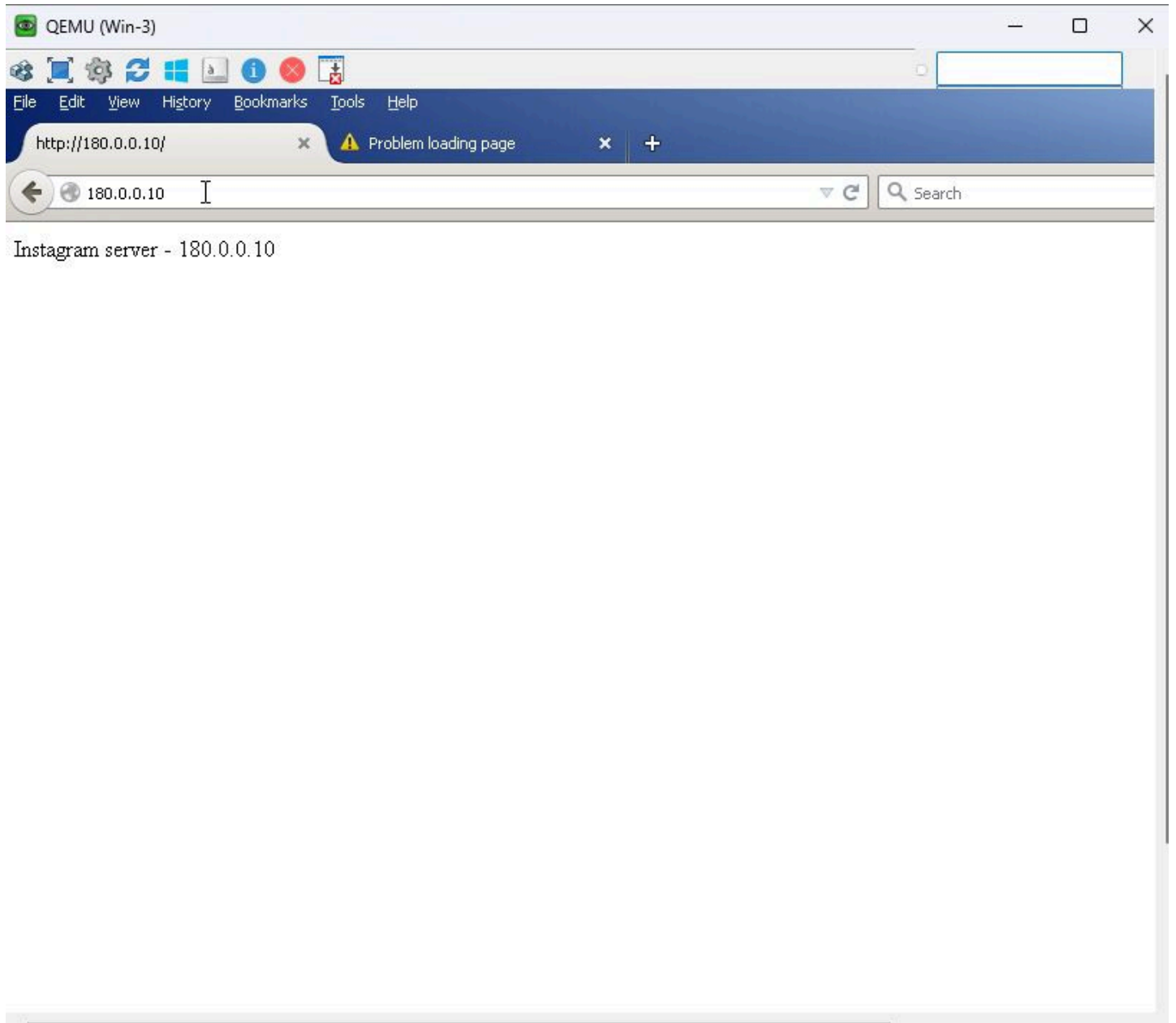
Pinging 192.52.234.100 with 32 bytes of data:

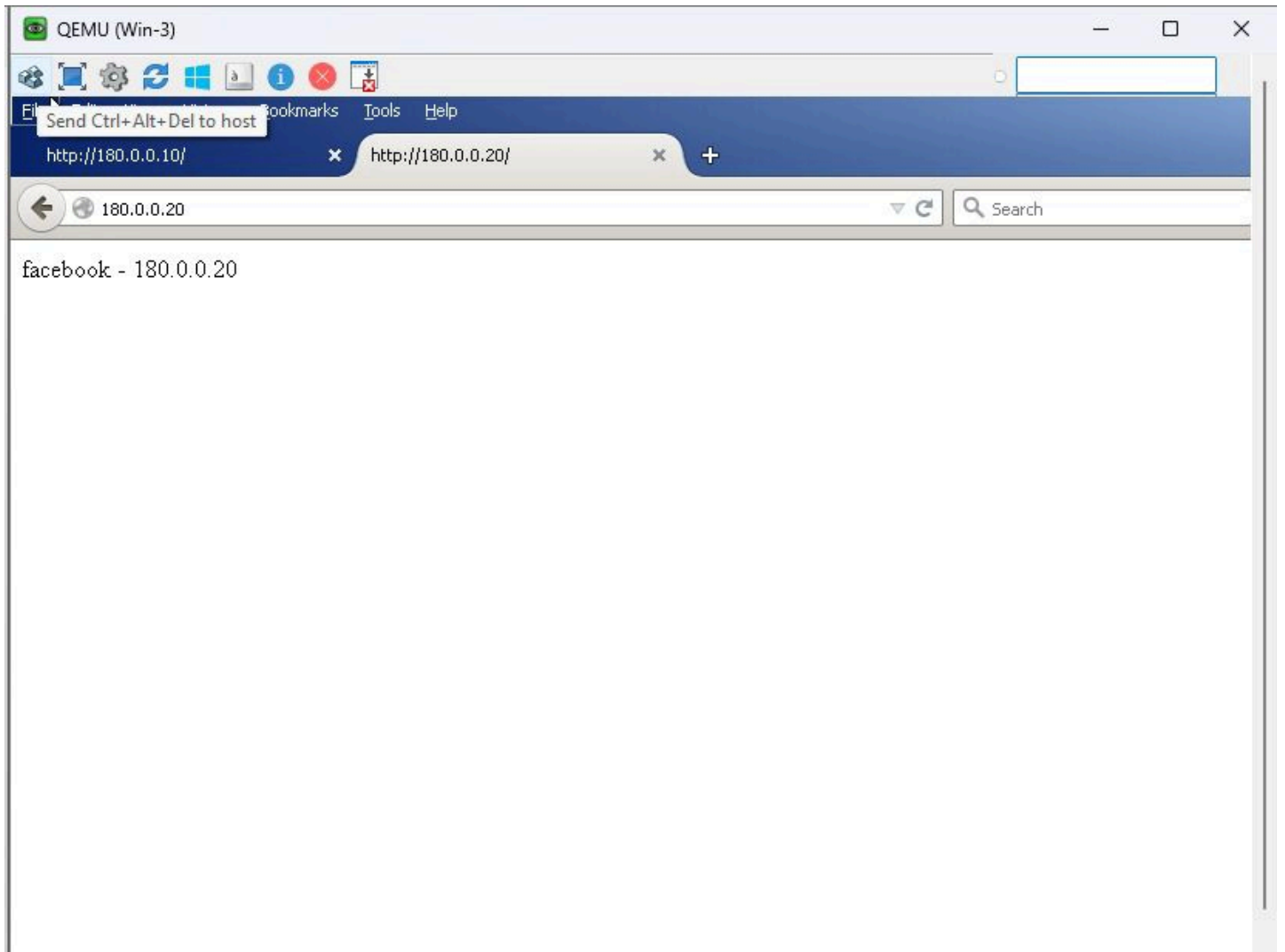
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

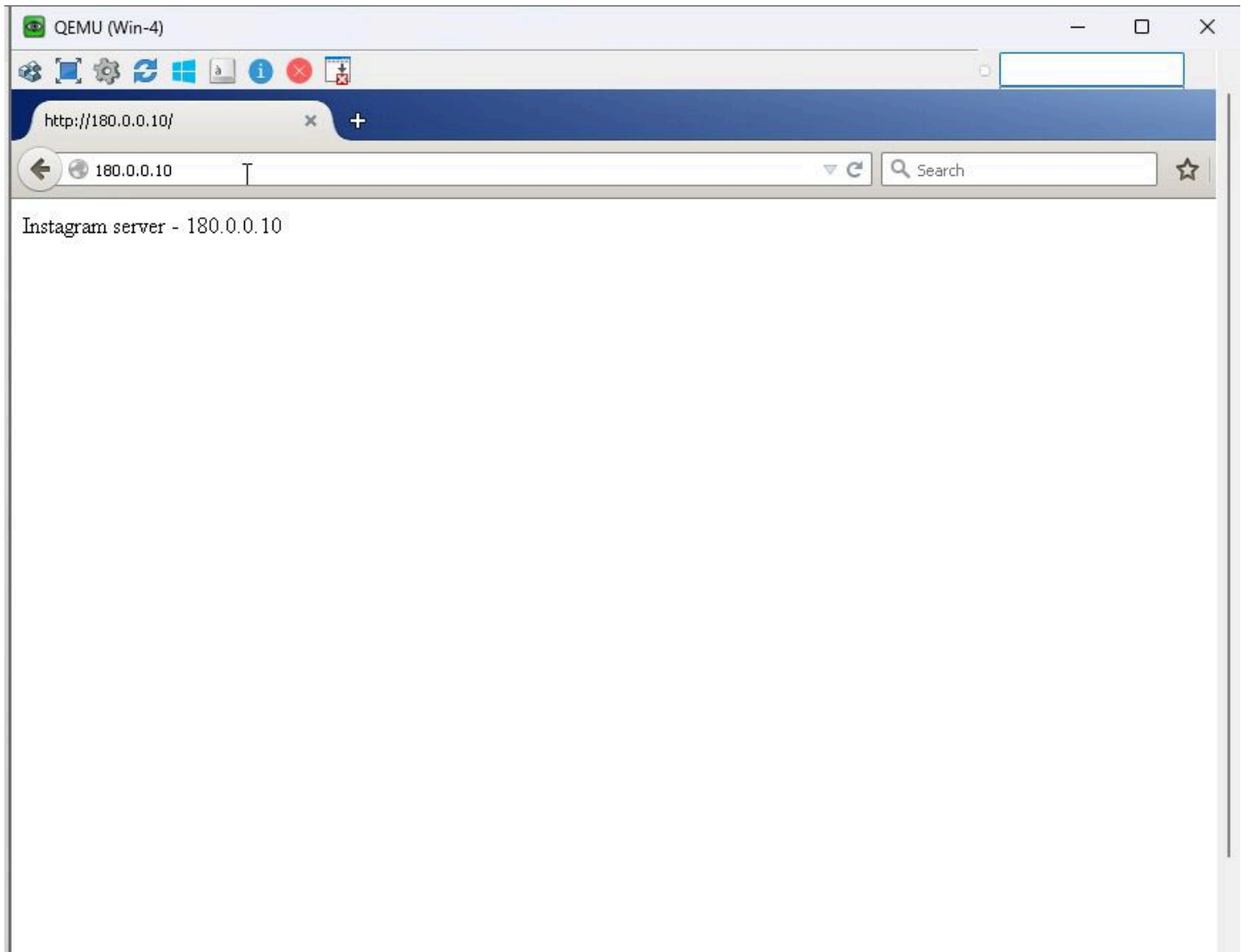
Ping statistics for 192.52.234.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

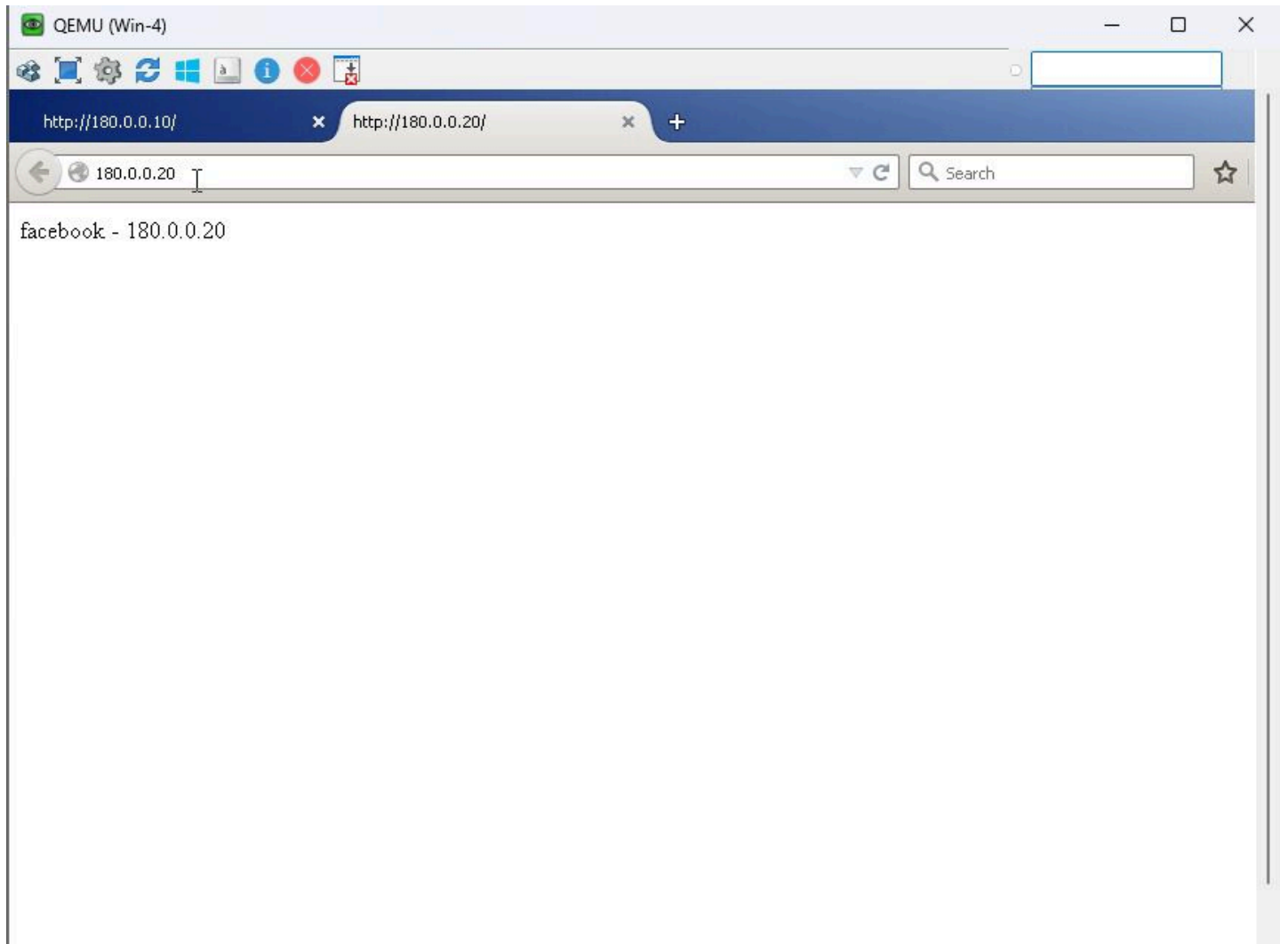
C:\Documents and Settings\user>_
```

```
R3
ip address 172.17.0.2 255.255.0.0
ip access-group 100 in
duplex auto
!
interface Ethernet0/2
ip address 172.18.0.1 255.255.0.0
ip access-group 1 in
duplex auto
!
interface Ethernet0/3
no ip address
ip access-group 100 in
duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
access-list 100 permit tcp host 10.0.0.10 host 192.52.234.200 eq www
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo-reply
access-list 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.0 0.0.255.255 eq www
!
control-plane
!
```









```
R3
!
interface Ethernet0/3
 no ip address
 ip access-group 100 in
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
access-list 100 permit tcp host 10.0.0.10 host 192.52.234.200 eq www
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo-reply
access-list 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.0 0.0.255.255 eq www
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo-reply
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo-reply
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 0 0
```



```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe

Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.

Ping statistics for 180.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 180.0.0.20

Pinging 180.0.0.20 with 32 bytes of data:

Reply from 180.0.0.20: bytes=32 time=2ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Reply from 180.0.0.20: bytes=32 time=3ms TTL=60

Ping statistics for 180.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>_
```

```
R3
!
interface Ethernet0/3
 no ip address
 ip access-group 100 in
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
access-list 100 permit tcp host 10.0.0.10 host 192.52.234.200 eq www
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo-reply
access-list 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.0 0.0.255.255 eq www
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo-reply
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo-reply
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 0 0
```

```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Reply from 180.0.0.20: bytes=32 time=3ms TTL=60
Ping statistics for 180.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\Documents and Settings\user>ping 192.52.234.100
Pinging 192.52.234.100 with 32 bytes of data:
Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.
Reply from 172.17.0.2: Destination net unreachable.
Ping statistics for 192.52.234.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\user>ping 192.52.234.200
Pinging 192.52.234.200 with 32 bytes of data:
Reply from 192.52.234.200: bytes=32 time=6ms TTL=60
Reply from 192.52.234.200: bytes=32 time=5ms TTL=60
Reply from 192.52.234.200: bytes=32 time=5ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 5ms
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
```

```
R3
duplex auto

interface Ethernet0/3
no ip address
ip access-group 100 in
duplex auto

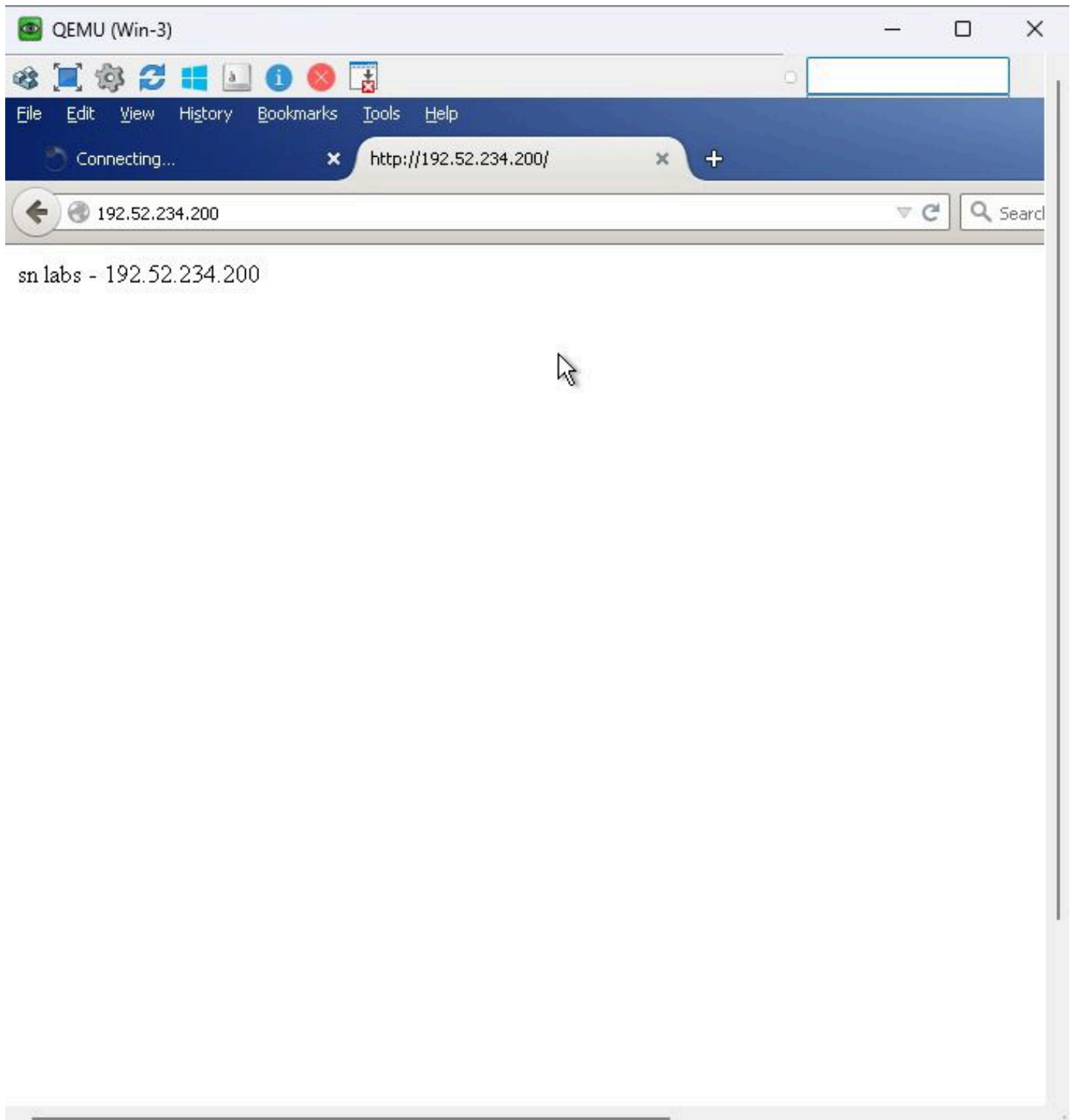
ip forward-protocol nd

no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2

ipv6 ioam timestamp

access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.52.234.100 eq www
access-list 100 permit tcp host 10.0.0.10 host 192.52.234.200 eq www
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 10.0.0.10 host 192.52.234.200 echo-reply
access-list 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.0 0.0.255.255 eq www
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo
access-list 100 permit icmp host 11.0.0.10 host 180.0.0.20 echo-reply
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo
access-list 100 permit icmp host 11.0.0.10 host 192.52.234.200 echo-reply
access-list 100 permit tcp 11.0.0.0 0.255.255.255 host 192.52.234.200 eq www

control-plane
```




```
R3
!
interface Ethernet0/0
 ip address 172.16.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/1
 ip address 172.17.0.2 255.255.0.0
 ip access-group 100 in
 duplex auto
!
interface Ethernet0/2
 ip address 172.18.0.1 255.255.0.0
 duplex auto
!
interface Ethernet0/3
 no ip address
 duplex auto
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 10.0.0.0 255.0.0.0 172.16.0.1
ip route 11.0.0.0 255.0.0.0 172.17.0.1
ip route 172.19.0.0 255.255.0.0 172.18.0.2
ip route 172.20.0.0 255.255.0.0 172.18.0.2
ip route 180.0.0.0 255.255.0.0 172.18.0.2
ip route 192.52.234.0 255.255.255.0 172.18.0.2
!
ipv6 ioam timestamp
!
!
access-list 100 deny      icmp host 10.0.0.0 host 11.0.0.0 echo
access-list 100 deny      icmp host 10.0.0.0 host 11.0.0.0 echo-reply
!
control-plane
!
!
!
```

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 11.0.0.10

Pinging 11.0.0.10 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 11.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 11.0.0.20

Pinging 11.0.0.20 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 11.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=128
Reply from 10.0.0.10: bytes=32 time=1ms TTL=128
Reply from 10.0.0.10: bytes=32 time=1ms TTL=128
Reply from 10.0.0.10: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 11.0.0.10

Pinging 11.0.0.10 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 11.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 11.0.0.20

Pinging 11.0.0.20 with 32 bytes of data:

Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.
Reply from 172.16.0.2: Destination net unreachable.

Ping statistics for 11.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>_
```